COMICCRAFT

AI Comic Story Creator using Gemini Models

AI-ML & GEN-AI Track - Complete Project Documentation

Date	30 September 2026
Team ID	XXXXXX
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	Track Documentation Marks

This document adapts the supplied AI-ML & GEN-AI Track project-template structure to the ComicCraft project described in the provided ComicCraft technical documentation. It preserves the template's phase-wise organization while filling the sections with ComicCraft-specific information.

1. BRAINSTORMING & IDEATION

Date	30 September 2026
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Maximum Marks	10 Marks

1.1 Brainstorming & Idea Prioritization

The brainstorming stage explores practical Generative AI ideas that can turn a simple user prompt into useful creative output. The final concept selected for this project is an AI-assisted comic creation platform that combines story generation and image generation in one workflow.

S.No	Idea / Suggestion	Category	Feasibility	Importance	Selected
1	AI Comic Story Creator	Generative AI / Creative	High	High	Yes
2	AI Social Media Caption Generator	Generative AI	High	Medium	No
3	AI Personalized Story Writer	Generative AI / NLP	High	High	No
4	AI Character Description Generator	Generative AI	High	Medium	No
5	AI Poster Generator	Generative AI / Vision	Medium	Medium	No
6	AI Children's Story Illustrator	Generative AI / Vision	Medium	High	No

Final selected idea: ComicCraft - a web application that accepts story preferences and automatically generates a five-panel comic with narration, dialogue, illustrations, preview, and PDF export.

1.2 Define Problem Statements

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Customer Problem Statement Template

PS	I am (Customer)	I am trying to	But	Because	Which makes me feel
PS-1	A student, storyteller, hobbyist, or non-artist	Create a complete comic from an idea	Creating artwork and writing every panel manually takes time and skill	Storytelling, dialogue, scene design, and illustration are separate tasks	Limited in expressing and sharing the idea
PS-2	A casual creative user	Iterate on tone and visual style	Changing the entire comic manually is difficult	Each panel needs coordinated text and visual prompts	Frustrated by repeated creative effort
PS-3	A project/demo user	Produce a shareable comic artifact	Text and images are often generated in separate tools	There is no single end-to-end workflow	Interrupted by tool switching

Problem Statement Summary

ComicCraft addresses the gap between a user's creative idea and a finished illustrated comic by coordinating prompt collection, structured story generation, image generation, preview, and PDF export through one FastAPI web application.

1.3 Empathy Map

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SAYS	THINKS	DOES	FEELS
“I have a story idea but cannot draw it.” “Can I make it funny or dramatic?” “I want all panels to feel connected.”	Wants the AI to understand the prompt. Wants consistent characters and scenes. Wants a quick result.	Enters story prompt. Chooses character, setting, tone, and art style. Reviews panels and downloads PDF.	Curious during generation. Excited by visual output. Disappointed if AI output is inconsistent. Satisfied when the comic is exportable.

Persona: Creative learner / storyteller / hobbyist who wants to convert an idea into a visually engaging comic without needing professional illustration skills.

2. REQUIREMENT ANALYSIS

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2.1 Customer Journey Map

Journey Stage	Actions	Touchpoints	Customer Thought	Customer Feeling	Opportunity
1. Idea Input	Enter story prompt, character, setting, tone and style	ComicCraft homepage form	“Will the system understand my idea?”	Curious / hopeful	Provide examples and validation
2. AI Generation	Submit form and wait for outline, story and images	FastAPI generation workflow	“Will the panels match my prompt?”	Anticipation	Show progress/status messaging
3. Review	Read each panel and inspect illustration	Comic preview page	“Does the story flow correctly?”	Engaged / evaluative	Keep panel structure clear
4. Export	Click PDF download	PDF export endpoint	“Can I save and share this?”	Satisfied	Provide reliable PDF generation

2.2 Data Flow Diagram

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Maximum Marks	2 Marks

ComicCraft data flow follows the supplied project architecture: the browser submits user preferences to FastAPI, the backend coordinates Gemini and image generation, then the layout builder and exporter create the final comic artifact.

USER / BROWSER		EXTERNAL AI SERVICES
Story prompt Character Setting Tone Art style	→	Gemini Flash Gemini Pro Stable Diffusion
ComicCraft FastAPI Backend	→	Generated outline / story / image
Layout Builder	→	Structured panel objects
FPDF Exporter	→	PDF in static/exports

Primary data stores/files: static/panels for generated illustrations and static/exports for generated PDF files.

2.3 Solution Requirements

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Functional Requirements (FR)

S.No	Requirement Category	ComicCraft Requirement	Priority
1	Input Processing	Capture story prompt, character name, setting, tone and art style through the web form.	High
2	AI Integration	Generate a structured five-panel outline using Gemini and expand it into narration/dialogue.	High
3	Image Generation	Generate comic-style illustrations from panel-specific image prompts.	High
4	Preview	Render generated panels with title, image, scene description, captions and narration.	High
5	Export	Compile panels and story content into a downloadable PDF.	High
6	JSON API	Accept structured JSON generation requests for programmatic clients.	Medium
7	Image Testing	Provide a route to test image generation independently.	Medium
8	Error Handling	Gracefully handle AI/API failures and malformed responses.	High

Non-Functional Requirements (NFR)

Category	Requirement	Acceptance Criteria
Performance	Generation should complete within practical limits for the selected AI services.	No server crash; clear status/error response.
Scalability	AI providers and model settings should be replaceable through configuration.	Service modules are separated from routes.
Security	API keys must be stored in environment variables.	No secret key hard-coded in source.
Reliability	Unexpected AI failures should be caught and reported.	HTTP error handling and safe fallbacks.
Usability	Simple, responsive form and sequential panel preview.	User can create and export a comic without technical knowledge.
Maintainability	Modular services for Gemini, image generation, layout and export.	Each major responsibility has a dedicated module.

2.4 Technology Stack

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S.No	Architecture Component	Technology Chosen	Justification / Purpose
1	Frontend / Client	HTML, CSS, Jinja2	Simple server-rendered interface with dynamic panel rendering.
2	Backend / Server	Python, FastAPI, Uvicorn	API routing, form/JSON handling, validation and ASGI serving.
3	AI Text Generation	Google Gemini Flash / Pro	Structured comic outlines, narration and dialogue.
4	AI Image Generation	Stable Diffusion / Hugging Face Diffusers	Comic-style illustrations from panel prompts.
5	PDF Generation	FPDF	Create multi-page downloadable comic PDFs.
6	Image Processing	Pillow	Image saving and preparation for export.
7	Configuration	python-dotenv / environment variables	Secure API-key and model configuration.
8	Version Control	Git / GitHub	Source control and collaboration.

3. PROJECT DESIGN PHASE

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Maximum Marks	15 Marks

3.1 Problem-Solution Fit

Canvas Area	ComicCraft Content
Customer Segment(s)	Students, hobbyists, non-artists, storytellers, educators and demo/project users.
Problems / Pains	Difficulty writing a complete comic, drawing panels, maintaining visual consistency, and exporting a finished artifact.
Triggers to Act	Need to quickly turn an idea into a comic, demonstrate Generative AI, or create a shareable story.
Emotions Before / After	Before: uncertainty and creative limitation. After: engagement, satisfaction and ability to share the result.
Available Solutions / Pros & Cons	Manual drawing tools and separate AI text/image tools provide control but require multiple workflows and more effort.
Customer Limitations	Limited drawing skill, limited time, and possible limits on AI API quotas or local compute.
Behavior	Prompt-driven creation, iterative style/tone changes, panel-by-panel review and PDF download.
Channels	Web browser, local FastAPI application and API endpoints.
Problem Root / Cause	Story writing, dialogue, illustration and document assembly are disconnected tasks.
Your Solution	One coordinated pipeline that transforms structured user input into a five-panel illustrated comic and PDF.

3.2 Proposed Solution

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Maximum Marks	5 Marks

Project Overview: ComicCraft is a web-based Generative AI application that transforms user-provided story information into a panel-by-panel comic with illustrations and downloadable PDF output. The supplied project documentation describes Gemini Flash for structured outlines, Gemini Pro for detailed narration/dialogue, and Stable Diffusion for illustrations. ■filecite■turn0file0■L101-L110■

Objective: Provide an end-to-end creative workflow from prompt entry to illustrated comic export.

Scope: Single comic generation with five panels, preview, image generation and PDF export. The source documentation identifies future enhancements such as user profiles, comic libraries and multi-page story arcs. ■filecite■turn0file0■L498-L510■

Core Features

• Story prompt and personalization inputs.

• Structured five-panel outline generation.

• Detailed narration and character dialogue.

• Comic-style illustration generation for each panel.

• Panel-by-panel web preview.

• Downloadable multi-page PDF.

• JSON API endpoint for programmatic generation.

• Independent image-generation test endpoint.

Resource Requirements

Resource	Specification / Allocation
CPU	Modern multi-core CPU suitable for Python/FastAPI development.
RAM	8 GB or higher recommended for application development; more is useful for local diffusion.
Storage	Sufficient space for Python environment, model caches, generated images and PDFs.
Software	Python, FastAPI, Uvicorn, Jinja2, Gemini SDK/API access, Diffusers/PyTorch, FPDF, Pillow.

3.3 Solution Architecture

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The supplied architecture diagram describes a browser-to-FastAPI flow, followed by HTML/Jinja2 rendering and AI services, then a layout builder, comic exporter and success page. ■filecite■turn0file0■L41-L42■

Presentation Layer Browser / HTML / CSS / Jinja2
↓↑
Application Layer FastAPI + Uvicorn routes.py / validation / workflow orchestration
↓↑
AI Integration Layer Gemini Flash → Outline Gemini Pro → Story / Dialogue Stable Diffusion → Images
↓↑
Processing / Export Layer layout_builder.py → exporters.py → PDF
↓
Storage static/panels + static/exports

Component	Role	Technology
Presentation	Collect inputs and render generated panels.	HTML, CSS, Jinja2
API / Routing	Receive form/JSON requests and coordinate workflow.	FastAPI, Uvicorn
Outline Generator	Create structured five-panel outline.	Gemini Flash
Story Generator	Create narration and dialogue.	Gemini Pro
Image Generator	Create panel illustrations.	Stable Diffusion / Diffusers
Layout Builder	Match panel stories with generated images.	Python
Exporter	Create downloadable PDF.	FPDF

4. PROJECT PLANNING PHASE

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4.1 Product Backlog, Sprint Schedule and Estimation

Sprint	Functional Requirement	User Story / Task	Story Points	Priority	Team	Output
1	Model & Architecture	Research Gemini and image generation models; define architecture.	5	High	Team	Architecture + model selection
2	Core AI	Implement outline and story generation services.	8	High	Team	Gemini workflow
3	Image Generation	Implement image generation and file storage.	8	High	Team	Panel images
4	Backend	Implement FastAPI routes, validation and error handling.	8	High	Team	Working API
5	Frontend	Build Jinja2 input, preview and success pages.	5	High	Team	Web UI
6	Export	Implement layout builder and PDF export.	5	High	Team	Downloadable PDF
7	Testing	Test mock mode, API routes, image generation and export.	5	Medium	Team	Test evidence
8	Documentation & Demo	Prepare documentation, screenshots and demonstration flow.	3	Medium	Team	Final submission

5. PROJECT DEVELOPMENT PHASE

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Maximum Marks	15 Marks

5.1 Coding & Solution

Field	ComicCraft Details
Repository / Source	Local Git repository / project folder; add GitHub or GitLab URL when published.
Programming Languages	Python, HTML, CSS, JavaScript (where used).
Frameworks	FastAPI, Uvicorn, Jinja2.
AI / ML	Gemini Flash, Gemini Pro, Stable Diffusion, Hugging Face Diffusers.
PDF / Image	FPDF, Pillow.
Key Features	Five-panel outline, story/dialogue, image generation, preview, JSON API, PDF export.
Setup	Create virtual environment, install requirements, configure .env, run uvicorn app.main:app --reload.

Development Workflow

The source documentation defines the core workflow as: receive user input → generate outline → generate story → generate images for each panel → build layout → save PDF → render preview/export success.
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5.2 Code Layout, Readability and Reusability

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S.No	Code Quality Parameter	ComicCraft Status	Remarks
1	Consistent Indentation	Yes	Python modules follow standard formatting conventions.
2	Proper File Structure	Yes	app, templates, static and configuration are separated.
3	Meaningful Variable Names	Yes	Names reflect prompts, panels, stories, images and exports.
4	Function / Method Names	Yes	Functions such as generate_outline, generate_story and save_pdf describe their purpose.
5	Code Comments	Partial	Add comments where AI parsing, fallbacks and export logic are non-obvious.
6	Modular Design	Yes	AI, image, layout and export responsibilities are separated.
7	No Redundant Code	Yes	Shared workflow helpers should be reused rather than duplicated.
8	Error Handling	Yes	AI and JSON failures should be caught and converted into user/API errors.

Component / Module	Language / Technology	Where Reused	Reusability
Gemini generation service	Python / Gemini API	Outline and story generation workflows	High
Image generator	Python / Diffusers	Every panel image	High
Layout builder	Python	Preview and PDF preparation	High
PDF exporter	Python / FPDF	Comic export	High
Jinja2 templates	HTML / Jinja2	Input, preview, success pages	Medium

5.3 No. of Functional Features Included in the Solution

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Maximum Marks	5 Marks

S.No	Feature Name	Module / Component	Status	Marks Contribution
1	Story Prompt Input	index.html / routes.py	Done	Core
2	Five-Panel Outline	Gemini Flash service	Done	Core
3	Narration & Dialogue	Gemini Pro service	Done	Core
4	Panel Illustration	Image generator	Done	Core
5	Comic Preview	comic_preview.html	Done	Core
6	PDF Export	layout_builder.py / exporters.py	Done	Core
7	JSON API	/generate-comic/json	Done	Additional
8	Image Test Endpoint	/test-image	Done	Additional

Feature Summary	Count / Value
Total Features Planned	8
Total Features Implemented	8 (documented target implementation)
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	To be recorded with final execution evidence

6. PROJECT TESTING

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Maximum Marks	5 Marks

6.1 Performance Testing

Performance testing should be executed separately for the web application and for AI-provider calls because model generation latency depends on external services and selected models.

S.No	Test Scenario	Virtual Users	Duration	Expected Outcome
1	Homepage load	1	30 sec	Page loads without server error.
2	Health/API check	5	60 sec	Stable response from FastAPI.
3	Mock comic generation	1	60 sec	Complete five-panel workflow succeeds without external AI.
4	Real AI generation	1	As available	AI response, image creation and PDF export complete or show graceful error.

Metric	Target	Actual	Status	Remarks
Average response time	< 2 sec for non-AI pages	Record during test	Pending evidence	AI generation is expected to be longer.
Maximum response time	< 5 sec for simple API pages	Record during test	Pending evidence	External model calls excluded from fixed target.
Throughput	Measure for simple endpoints	Record during test	Pending evidence	Use repeated health/API calls.
Error rate	< 1% for stable local endpoints	Record during test	Pending evidence	AI provider quota errors should be classified separately.
CPU utilization	< 80% under expected load	Record during test	Pending evidence	Local Diffusers may require higher compute.
Memory utilization	< 80%	Record during test	Pending evidence	Local image generation can be memory-intensive.

7. PROJECT DOCUMENTATION

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7.1 Project Executable Files

Item	Expected Location / Command	Purpose
Application entry point	app/main.py	Creates and configures the FastAPI application.
Routes	app/routes.py	Homepage, generation, JSON API, image test and export-success routes.
Gemini service	app/gemini_* or services module	Outline and story generation.
Image generator	app/image_generator.py or services module	Create and save panel illustrations.
Layout builder	app/layout_builder.py or services module	Combine story data and images.
Exporter	app/exporters.py or services module	Generate PDF files.
Templates	templates/*.html	Frontend pages.
Static output	static/panels and static/exports	Generated images and PDFs.
Environment	.env	API keys and configuration; do not commit secrets.
Requirements	requirements.txt	Python dependency installation.

Run command: **uvicorn app.main:app --reload**. Open **http://127.0.0.1:8000** and FastAPI documentation at **http://127.0.0.1:8000/docs**. The supplied project documentation specifies this local deployment flow. ■filecite■turn0file0■L419-L432■

7.2 Sample Project Documentation

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Maximum Marks	10 Marks

Project Summary

ComicCraft demonstrates an end-to-end Generative AI workflow that transforms a creative prompt into a structured five-panel comic, including story narration, character dialogue, scene descriptions, AI-generated illustrations and a downloadable PDF. The source documentation describes the system as a FastAPI web application integrated with Gemini and Stable Diffusion. ■filecite■turn0file0■L2-L15■

Documented User Flow

• User opens the ComicCraft homepage.

• User enters story prompt, main character, setting, tone and art style.

• FastAPI receives the form or JSON request.

• Gemini Flash generates a structured five-panel outline.

• Gemini Pro expands the outline into narration and dialogue.

• Stable Diffusion generates the panel illustrations.

• The layout builder associates images with panel stories.

• FPDF creates the multi-page comic PDF.

• The preview page displays the result and the export-success page confirms completion.

The supplied documentation identifies the homepage fields and the preview page contents in detail. ■filecite■turn0file0■L437-L472■

8. PROJECT DEMONSTRATION

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Maximum Marks	5 Marks

8.1 Project Demo Planning

S.No	Demo Section	Description	Duration	Responsible Member
1	Introduction & Problem	Explain manual comic creation problem and project objective.	2 min	Member 1
2	Architecture	Show FastAPI, Gemini, image generation, layout and export pipeline.	2 min	Member 2
3	Live Input	Enter prompt, character, setting, tone and art style.	2 min	Member 1
4	AI Output	Show generated five panels and narration/dialogue.	4 min	Member 2
5	PDF Export	Download and open generated comic PDF.	2 min	Member 3
6	Q&A;	Explain technologies, limitations and future scope.	3 min	All

Recommended demo sequence: introduction → problem statement → architecture → live comic creation → panel review → PDF download → Q&A.;

8.2 Demonstration of Proposed Features

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S.No	Feature Name	Description	Status	Demonstrated
1	Personalized Input	Prompt, character, setting, tone and style selection.	Implemented	Yes
2	AI Outline	Five-panel structured storyline.	Implemented	Yes
3	Narration & Dialogue	Detailed panel story content.	Implemented	Yes
4	AI Illustrations	Comic-style image for each panel.	Implemented	Yes / if provider available
5	Comic Preview	Sequential visual/text panel review.	Implemented	Yes
6	PDF Export	Download complete comic as PDF.	Implemented	Yes

Feature Implementation Summary	Value
Total Features Proposed	6 core demonstration features
Total Features Implemented	6 documented features
Total Features Demonstrated	Record actual demo evidence
Overall Implementation Rate	100% documented target; verify with final run

8.3 Scalability & Future Plan

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S.No	Current Limitation	Impact	Priority
1	AI provider quotas / rate limits	Generation can temporarily fail or be delayed.	High
2	Single-comic workflow	No persistent comic library or user history.	Medium
3	Image-generation compute dependency	Local generation can be resource intensive.	High
4	Character consistency	Separate panel prompts may produce visual variation.	Medium

Aspect	Current State	Proposed Upgrade / Solution
User Load	Single/local deployment	Deploy behind a scalable ASGI service and queue long AI jobs.
Data Storage	Static files	Object storage/database for comic metadata and assets.
Performance	Synchronous generation workflow	Background task queue, caching and progress updates.
Security	Environment-based secrets	Secret manager, authentication, authorization and rate limiting.
AI Quality	Independent panel prompts	Character-reference conditioning and consistency controls.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User accounts and comic history	Future iteration	Persistent personalized library.
Phase 3	Longer story arcs and more panels	Future iteration	Support multi-page comics.
Phase 4	Character consistency and richer layouts	Future iteration	Higher visual continuity and professional layouts.

8.4 Team Involvement in Demonstration

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Member	Responsibility	Demo Contribution
Member 1	Frontend / User Flow	Explain inputs, homepage and comic preview.
Member 2	AI / Backend	Explain Gemini workflow, image generation and API integration.
Member 3	Export / Testing	Demonstrate PDF export, testing and troubleshooting.
All Members	Q&A; / Documentation	Answer technical and project-planning questions.

8.5 Communication

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Maximum Marks	1 Marks

Communication Area	ComicCraft Practice
Daily / Sprint Updates	Share implementation progress, blockers and next tasks.
Technical Coordination	Document model configuration, API errors, prompt changes and test results.
Version Control	Use Git commits with clear messages and keep secrets outside the repository.
Demo Communication	Use a fixed narrative: problem → solution → architecture → live feature → result → future scope.
Issue Escalation	Record API quota, authentication, model availability and image-generation failures separately.

CONCLUSION

ComicCraft provides a complete Generative AI project workflow: personalized prompt input, structured comic outline generation, detailed storytelling, image generation, panel preview and PDF export. The supplied project documentation explicitly describes this end-to-end architecture and identifies future extensions such as profiles, comic libraries and multi-page story arcs. ■filecite■turn0file0■L498-L510■

Source basis: ComicCraft project documentation supplied in this conversation and the AI-ML & GEN-AI Track Project Template supplied in the latest upload. The track template's terminology and phase structure have been preserved while the content has been adapted to ComicCraft.